

HW06 - AI-assisted API Testing

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Course: Software Testing

SUT: EShop - <<https://github.com/ttbhanh/eshop-sut>>

Repository: <https://github.com/KieuDuyennn/software__testing/tree/hw6/HW06>

Latest execution date: 2026-08-24

1. Scope and API selection

The three graded pool selections are API 1 (Pool A), API 3 (Pool B), and API 4 (Pool C). API 2 / FR-06 is additional Pool-A coverage and is reported separately so it does not obscure compliance with the one-per-graded-pool rule.

API	Pool	Requirement	Endpoint	Role in submission
API 1	A	FR-01 Account registration	POST /api/register	Graded Pool A
API 2	A	FR-06 Product detail	GET /api/products/:id	Additional coverage
API 3	B	FR-11 Order history	GET /api/orders/my-orders, GET /api/orders/:id, cancellation	Graded Pool B
API 4	C	FR-13 Dashboard	GET /api/admin/orders, admin status transition	Graded Pool C

FR-13 has no dedicated dashboard endpoint. The frontend derives order count and revenue (sum of total_amount for status = delivered) from GET /api/admin/orders, so this is the tested API.

Group uniqueness confirmation (2026-08-24): I, Lê Phạm Kiều Duyên (23127184), confirm that I checked with my group and that the graded combination FR-01 Account registration (Pool A), FR-11 Order history (Pool B), and FR-13 Admin dashboard (Pool C) is unique within the group. API 2 / FR-06 is additional Pool-A coverage and is not part of this three-API uniqueness combination.

2. Test environment and reproducibility

Item	Value
Backend	Node.js + Express + SQLite, freshly reseeded per run
Base URL	http://localhost:3000
Runner	Postman collection format + Newman 6.2.1
Reports	Newman JSON + newman-reporter-htmlextra HTML
Mandatory identity	X-Student-Id: 23127184, injected and asserted at collection level
Rate limiting	LOADTEST=1 during suite runs to avoid invalid HTTP 429 noise
Full run log	evidence/newman-console/suite_full_20260824-002523.log
Gate run log	evidence/newman-console/suite_gate_20260824-002730.log

Reproduce locally:

```
npm install
npm run sut:install
.\scripts\Invoke-ApiTests.ps1
.\scripts\Invoke-ApiTests.ps1 -Mode gate
```

3. AI-first method and human-review control

Each API followed four traceable stages:

1. Generate partitions for every parameter, state transitions, SEC-01..SEC-07, and strict response schemas.
2. Audit every generated case as VALID, INVALID, or INCOMPLETE against the specification and requirements, not against observed SUT behavior.
3. Add five student-designed cases targeting blind spots found in the audit.
4. Execute on a fresh database, then map failures to root-cause bugs rather than treating every failed assertion as a separate defect.

One generated case was INVALID: A2-DP-006 duplicated product-id 5. It was replaced with a percent-encoded-id case. Twenty-two cases were INCOMPLETE because the specification could justify only a weak safety oracle or omitted the behavior entirely. The remaining 363 final cases were VALID. Full reasoning for all 386 rows is in docs/phases/*/02-audit.md.

4. Results summary

API	AI-generated	Student-added	Total	Fully passed	Failed cases	Assertions passed	Assertions total
API 1 - FR-01	121	5	126	69	57	544	606
API 2 - FR-06	83	5	88	54	34	390	433
API 3 - FR-11	89	5	94	84	10	407	418
API 4 - FR-13	73	5	78	68	10	333	345
Total	366	20	386	275	111	1,674	1,802

The 128 failed assertions are expected defect-revealing results. They are clustered into 16 confirmed bugs. The separate regression gate contains cases that the current SUT satisfies and passed 1,262/1,262 assertions.

5. API 1 - FR-01 Account registration

Generation produced 121 cases across name/email/password domains, registration state, SEC controls, and response schemas. Audit found ambiguous limits and one misattributed security case: A1-SEC-013 originally blamed injected role data, but A1-SEC-012 proves that field is ignored. It was corrected into a direct admin-authorization test.

Student additions cover tab/CRLF email controls, NUL in name, JSON media type with charset, and numeric confirmPassword. Three of these produced new evidence for the existing validation root cause. The run confirmed BUG-01, BUG-02, BUG-06, and BUG-07 through BUG-12. Detailed registers:

- docs/phases/api1-fr01-register/02-audit.md
- docs/phases/api1-fr01-register/03-extend.md
- docs/phases/api1-fr01-register/04-execute.md

6. API 2 - FR-06 Product detail (additional)

The corrected final suite has 88 cases. Added cases test URL-decoded valid ids, path/query precedence, explicit JSON content negotiation, double encoding, and full-width Unicode digits. The run confirms three root causes: unknown or malformed ids return 200 {} (BUG-03), even ids expose price as a string (BUG-04), and product write routes lack authentication/role checks (BUG-13).

Detailed registers are under docs/phases/api2-fr06-product-detail/.

7. API 3 - FR-11 Order history

The 94-case suite uses real two-user fixtures and walks the complete FR-10 state machine. Student additions emphasize post-conditions and cross-route consistency. A3-HR-001 shows that the forbidden shipping cancellation not returns the wrong status and changes the order to canceled.

Confirmed root causes: anonymous/cross-user order detail IDOR (BUG-05), missing admin role checks (BUG-06), deleted-account JWTs remain usable (BUG-14), shipping cancellation is accepted (BUG-15), and canceled -> delivered is accepted (BUG-16).

Detailed registers are under docs/phases/api3-fr11-order-history/.

8. API 4 - FR-13 Admin dashboard

The 78-case suite validates authorization, revenue/count projections, every legal and illegal order transition, and the joined response schema. Added cases test aggregation additivity, unique order ids, same-state terminal replay, authorization atomicity, and one-checkout/one-row count integrity.

A4-HR-004 independently proves a user token can make a legal admin transition and mutate confirmed -> shipping. A4-ST-016 connects the invalid canceled -> delivered transition to false delivered revenue. These failures map to BUG-06 and BUG-16.

Detailed registers are under docs/phases/api4-fr13-admin-orders/.

9. Postman features

Verified locally: collections, nested folders, local/CI/mock environments, environment and global variables, secret variable metadata, collection- and request-level pre-request scripts, pm.sendRequest fixture chains, token capture, strict JSON-schema assertions, dynamic values, Newman CLI, HTML/JSON reporting, selective regression gates, and CI configuration.

The signed-in Postman workspace contains all four HW06 collections and the EShop - Local (23127184) environment; both views are captured under evidence/postman-cloud/. A real Postman Desktop request to http://localhost:3000/api/products/2 returned HTTP 200, and its Console image shows [HW06] X-Student-Id=23127184. A public API2 Mock Server is also running and has recorded a real call. Collection Runner, Monitor and a successful mock example response remain pending.

10. CI/CD

The workflow separates a blocking green regression gate from a non-blocking full defect-discovery suite. The local equivalent is verified green at 1,262/1,262. The initial pushed run is authentic red evidence caused by two Linux-unstable stateful cases; the reviewed follow-up gate passed green in GitHub Actions. Both SHAs and run URLs are in docs/cicd/CI_CD_REPORT.md.

11. Bug summary

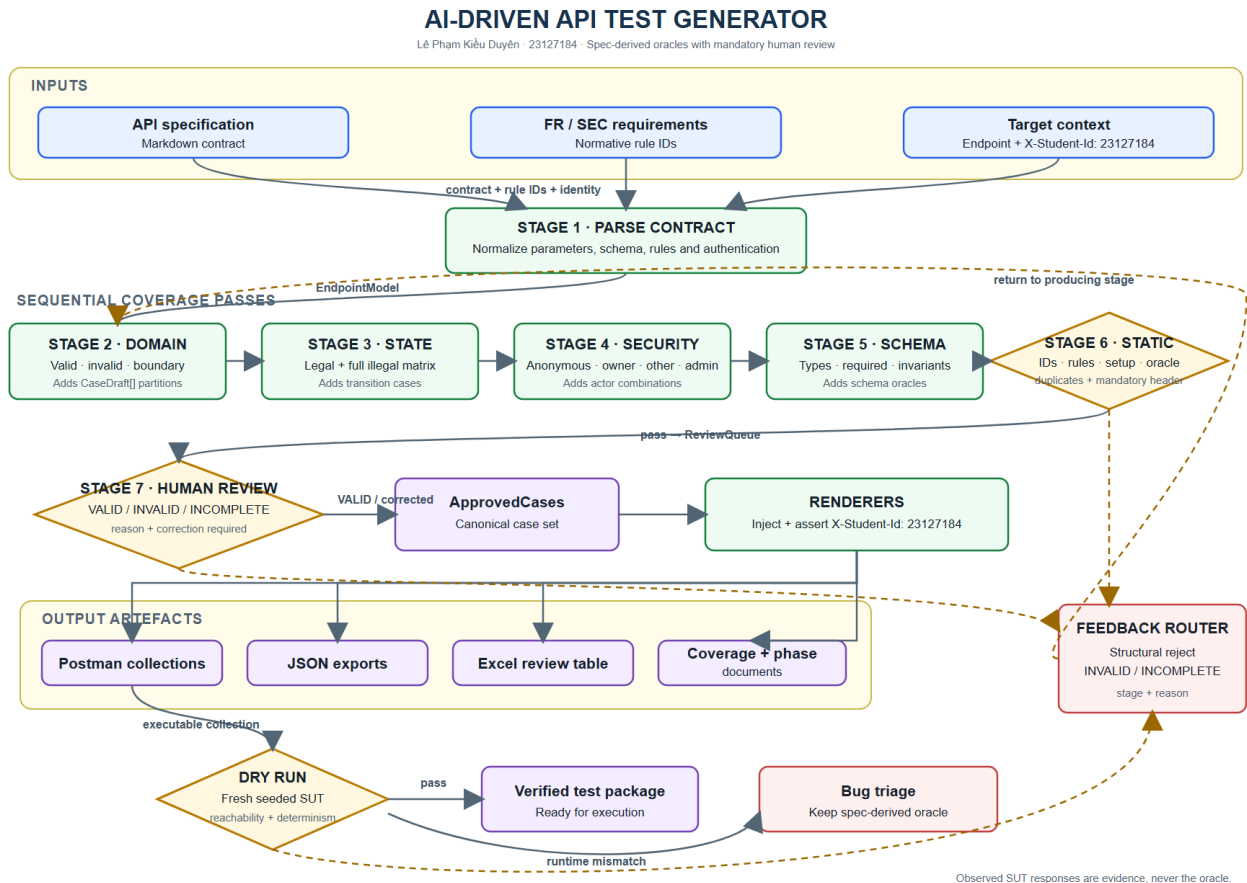
Range	Count	Main themes
BUG-01..BUG-12	12	Registration validation, plaintext passwords, product response contract, IDOR, admin authorization, error handling
BUG-13..BUG-16	4	Product write authorization, stale JWT, invalid cancellation, terminal-state violation
Total	16	Full reproduction steps and Newman messages in docs/bugs/BUG_REPORT.md

Real GitHub Issue URLs are recorded in the bug report. Signed-in issue-page screenshots for #66-#70 and real green/red Actions summaries are stored under evidence/screenshots/; evidence/EVIDENCE_INDEX.md maps every image to its visible verification fields.

12. AI-driven test generator (G9.5)

The generator uses a normalized endpoint model, four coverage passes, mandatory human review, executable rendering, a dry-run validator, and a feedback loop. Design and pseudocode are in docs/design/. A reusable implementation is under .claude/skills/api-test-generator/.

The generator flow is exported at docs/design/diagram/generator-design.png, with editable Mermaid source beside it. It shows the three contract inputs, seven design stages, mandatory VALID / INVALID / INCOMPLETE human-review gate, rendered artifacts, dry-run validation, bug triage, and feedback paths. The current diagram is an editable draft. It requires final student review and explanation in the narrated demonstration video.



AI-driven API test generator pipeline

13. AI Critique

See docs/AI_CRITIQUE.md (249 words). It identifies the duplicated case, misattributed security failure, and missing post-condition oracles, and derives the rule that generated cases require human approval before they become test evidence.

14. AI Audit and evidence integrity

The mandatory interaction log is in docs/ai-audit/AI_AUDIT.md. Local Newman JSON, HTML, and timestamped console logs are real execution artifacts. The following are intentionally pending because they require the student/device: Collection Runner, Monitor, a successful mock example response, final student authorship review of the editable diagram, and the narrated YouTube video. The mandatory executed Postman Console evidence is complete. Group uniqueness was confirmed by the student on 2026-08-24.